Proposal for Revising the Undergraduate Curriculum Department of Physics and Astronomy

October 10, 2025

1 History and Status of the Proposal

This proposal has been developed over more than five years within the Department of Physics and Astronomy, led by the undergraduate curriculum committee. The proposal was subjected to an extensive vetting process that involved assigned department readers who were not involved in formulating the initial plan. The proposal was unanimously supported by the department in a December 2021 vote. All of the new courses, and courses with major content modification, have been approved in the UC Davis Integrated Curriculum Management System (ICMS). The discontinued courses will not be dropped from the catalog until the new major requirements are approved.

The impact of the proposal outside of the Department of Physics and Astronomy is expected to be minimal. The proposal leaves PHY 7 and 9 unchanged, apart from a long overdue clean up of the mathematics prerequisites, which make them consistent with the equivalencies accepted by the Department of Mathematics, and increases the options available to students. The proposal does eliminate PHY 9HE, but a search of the current course catalog does not indicate any majors that explicitly list PHY 9HE as a requirement (even an alternate) or a prerequisite.

Our revisions to the computational curriculum involves the replacement of PHY 102 and PHY 104B with PHY 45, 110L, 112L, and 115L. The third quarter of electricity and magnetism, PHY 110C is also eliminated, with the mathematical review that used to be part of the sequence taken up in PHY 9 and PHY 104. Details concerning how these affect each specific major in the program are found in the corresponding sections 4-6. (A third quarter of quantum physics, PHY 115C, will eventually be added as an elective, partly in recognition of the rapid development and growth of quantum information science, but this change is several years in the future and hence not currently in ICMS.) The upper division physics instrumentation sequence PHY 116ABC will evolve into PHY 117 and 118 with PHY 80 as a prerequisite. None of these changes will affect majors outside of physics, nor will they necessitate any additional instructors.

In 2024 and 2025, the proposal was updated to adjust to changing prerequisites and course availability, which pushed some applied physics majors over the unit caps, after accounting for new implicit prerequisites. In this updated proposal, there are no implicit prerequisites: if a course is needed for a required course, it is included as a required course. The unit tallies for all majors have been checked several times, and clarifying remarks and tallies have been added. The availability and pre-requisites of all courses was also confirmed in the catalog. We also updated the proposed implementation, to reflect the fact that many of the new proposed required courses have already been approved and taught. Students have been encouraged to take advantage of the new offerings, even though they are not yet required courses for the major.

2 Objectives of the Proposal

Table 1: An example schedule satisfying the *current requirements* for an undergraduate physics major that takes the 9H series and 122A. Physics course numbers are shown with the number of units in parenthesis. Math courses start with M. Courses in italics are electives or have at least two different offerings per year. Course CAP is a capstone elective, and course X is any advanced elective.

Year	Fall	Winter	Spring
Freshman	9HA(5) *M21B(4)*	9HB(5) *M21C(4)*	9HC(5) *M21D(4)*
Sophomore	9HD(5) *M22A(3)*	9HE(5) *M22B(3)*	40(3) *80(4)*
Junior	104A(4) 105A(4) 102(1)	105B(4) 110A(4)	110B(4) 115A(4)
Senior	110C(4) 112(4) 115B(4)	*122A(4)* *CAP(4)* *CAP(4)*	*X(3-4)* *X(3-4)* *CAP(4)*

The changes proposed here reflect three main observations:

[i.] Workload over four years of study was imbalanced, with too few physics courses in the sophomore year, and too many in the junior year.

[ii.] Integration between transfer students and four-year students, who have somewhat different backgrounds, needed to be improved. Our recent departmental Climate Survey confirmed this by showing a huge satisfaction gap between the groups.

[iii.] The curriculum did not reflect the explosive growth in computational methods and their applications.

An example schedule of student coursework without the revisions proposed here is shown in Table 1 for students taking honors physics. An example schedule for students that transfer to UC Davis for their junior year is shown in Table 2. There are many different trajectories through our program, but most are some variation on these two. For consistent comparisons, all the example schedules for BS Physics majors in this proposal assume the students take 122A and one capstone course in the winter, and two capstone courses in the spring. Many other lab courses and offerings are possible.

There are some deficiencies in the current course of study:

- Physics majors who complete 9HD or 9D in the fall of their sophomore year have little to do for the rest of the year. The honors sequence has 9HE, but that course does not contain any content which is a prerequisite for upper division class work. The recent addition of 40 and 80 is helpful in that it provides something for students to do during this time, but the problem still remains that they do not make progress on the upper division core coursework, and 80 is not taken by all students. The result of this stalling is that the junior and senior years are a race to complete the degree requirements, leaving very little flexibility or time for advanced electives.

Table 2: An example schedule satisfying the *current requirements* for an undergraduate physics major that transfers to UC Davis in their junior year and takes 122A. Most of our incoming transfer students have not taken an equivalent course for at least one of PHY 9D, 40, or 80. This example considers the extreme case where they have completed none of these courses upon arrival in the junior year. Physics course numbers are shown with the number of units in parenthesis. Courses in italics are electives or have at least two different offerings per year. Course CAP is a capstone elective, and course X is any advanced elective.

Year	Fall	Winter	Spring
Junior	9D(4) 104A(4) 105A(4) 102(1)	105B(4) 110A(4) *80(4)*	40(3) 110B(4) 115A(4)
Senior	110C(4) 112(4) 115B(4)	*122A(4)* *CAP(4)* *CAP(4)*	*X(3-4)* *X(3-4)* *CAP(4)*

- Students that transfer to UC Davis for their junior year face a wall of coursework that they have to handle in the first quarter: math methods, mechanics, and modern physics. Many also take 102, which instructors have found challenging to teach within the workload limits of a one-unit course. For many students, these are also the first courses they encounter that require solving challenging homework problems. We have two trains of students running through our program and the fall of their junior year is the train wreck where they collide.

- The current curriculum does not include sufficient computing practice for our students. It is useful to consider what a physics degree would look like if we taught calculus the same way we teach computing. Students would arrive their freshman year and take an introductory calculus course. Then, they would take their physics courses, which would never mention calculus. At some point in their junior or senior year, they would take a one quarter course called "Calculus in Physics" which would attempt to show all the ways we use calculus in physics. Our students are experts at calculus because they learn how to use the tool, and then apply it, again and again, throughout their coursework. To remain relevant in the modern world (or even the world from 20 years ago) our majors need more practice in the use of computing as an essential tool for solving physics problems.

- Four-year students typically graduate with around 180 units. Within the College of Letters and Science, we are allowed to require a maximum of 110 units in our majors. The current BS physics major requires a minimum of 108 units to complete. Fitting the canon of undergraduate physics into such a tight space is extremely challenging. Students complain that we waste time teaching some topics again and again (e.g. Special Relativity from scratch) while completely dropping other topics (e.g. Classical Hamiltonians). The problem is particularly acute for applied physics majors, where core material must be dropped to make space for coursework outside of physics.

- The prerequisite structure of the upper division courses creates many tiers. As an extreme example, 122 requires 112, which requires 115A, which requires 104A and 105A, both of which require the 9 series. This, combined with the rapid pace, leaves very little flexibility for students once they start their junior year. For example, missing any of the four-unit courses in the junior year of Table 1 requires an exception to prerequisites or an extra year to graduate.

This proposal aims to make significant improvements on these issues, focused on the following considerations:

- Use the sophomore pause more productively, lower the junior year brick wall.
- Provide more practice and training in computational physics
- Our niche: designing, programming, and validating numerical models and algorithms.
- Computing and analytic skills are complementary, it's not one vs the other.
- Avoid reteaching and missed topics.
- Simplify prerequisite structure and reliably deliver prerequisite material.
- We must preserve a two year graduation path for students that transfer in their junior year.
- We cannot exceed 110 units of required coursework (including math).

3 Proposed BS Requirements

The proposed required courses for a BS in physics are presented in Tables 3 and 4. The math prerequisites for lower division coursework are broken out separately in Table 6. Example schedules are presented in Tables 7-9.

A primary feature of this proposal is that incoming transfer students now overlap in some courses with sophomores who took the honors physics sequence. This eliminates the stalling of our four-year students while relieving some of the intense academic pressure on incoming transfer students. Our experience has been that the best of our transfer students perform as well as the best of our four-year students, once they have sufficient time to adjust, and this proposal gives them that time. Accelerating students who took the honors sequence also provides tremendous additional flexibility to their schedule. The example schedule in Table 7 assumes the student takes each class nearly as soon as possible, leaving ample time in their senior year for additional electives.

Transfer students who wish to complete their degree in two years are still highly constrained, but instead of facing three upper division courses immediately upon arrival they now start with one upper division course. Take the time to compare fall quarter of the junior year in Tables 2 and 9; this is a major feature of this proposal. This gentle introduction does not come at the cost of dramatically increased unit loads later on: transfer students can still complete the degree without exceeding 13 units of physics coursework in any quarter.

The college imposes a limit of 110 units of required coursework, including prerequisites, for any major. Within a particular major, options that exceed this limit are permitted, as long as a path that stays below the limit is available. The current BS physics major requires a minimum of 108 units. This proposal takes advantage of the remaining two units and brings the minimum to 110 units. This slightly increases total unit pressure on students, which particularly affects those transfer students who have only two-years to absorb them. However, the proposal replaces the one-unit course PHY 102 with the more appropriately sized four-unit PHY 45 course. Also, the major stress point for transfer students is in their first quarter, which this proposal substantially improves. The proposal also adds significantly more schedule flexibility, which should also help alleviate pressure.

Table 3: Preparatory Subject Matter (All Physics Majors)

Units (minimum): 52 (48 when PHY 45 is not required)
See Table 6 for math prerequisites.

Course		Units	Offered	Prereqs	Name
MAT	21A	4	FWS		Differential Calculus
	21B	4	FWS		Integral Calculus
	21C	4	FWS		Partial Derivatives and Series
	21D	4	FWS		Vector Analysis
				One of:	
MAT	22A†	3	FWS		Linear Algebra
	27A	4	FWS		Linear Algebra
	67	4	FWS		Mod. Linear Algebra
				One of:	
MAT	22B	3	FWS		Differential Equations
MAT	27B	3	FWS		Differential Equations
				Either:	
PHY	9A	5	FWS		Class. Physics *(Class. Mech.)*
	9B	5	FWS	9A/9HA(C-)	Class. Physics *(Waves, Thermo, Optics)*
	9C	5	FWS	9A/9HA, 9B(C-)	Class. Physics *(Elec. and Magn.)*
	9D	4	FS	9B(C-), 9C(C-)	Mod. Physics *(Rel. and QM)*
				or	
PHY	9HA	5	F		Honors Physics *(Class. Mech.)*
	9HB	5	W	9HA(C-) or 9A(B-)	Honors Physics *(Rel. and Stat. Mech.)*
	9HC	5	S	9HA(C-) or 9A(B-), 9HB(C-)	Honors Physics *(Waves and QM)*
	9HD	5	F	9HB(C-), 9HC(C-)	Honors Physics *(Elec. and Magn.)*
PHY	40	3	F		Intro. to Physics Computation
	45‡	4	W	40(C-), 9C/9HD(C-)	Computational Physics
	80	4	WS	9C/9HD(C-), 40(C-)	Experimental Techniques
	185*	1	S		Careers in Physics
	190*	1	F		Careers in Physics

*: recommended.

†: MAT 22A has a co-requisite of 22AL (or other options) which so far is not being enforced; we are asking if PHY 40 can cover it instead.

‡: **PHY 45 is not required for the Astrophysics, Computational Physics, and AB degrees.**

Table 4: Depth Subject Matter (BS Physics)

Units (minimum): 39

Course		Units	Offered	Prereqs	Name
PHY	104A	4	FS	9C/9HD(C-),‖40 MAT 21D(C-),22A/27A/67(C-), MAT 22B/27B(C-)	Mathematical Physics
	105A	4	W	9A/9HA(C-),9C/9HD(C-), MAT 22A/27A/67(C-), MAT ‖22B/‖27B	Classical Mechanics I
	105B	4	S	104A(C-),105A(C-)	Classical Mechanics II
	110A	4	W	104A(C-), 9C/9HD(C-)	Electricity and Magnetism I
	110B	4	S	104A(C-), 110A(C-), 9D/9HB(C-)	Electricity and Magnetism II
	110L	1	S	40(C-), 45/ECS 36B/32C(C-) 110A, ‖110B	Comp. Lab in Elec. and Magn.
	112	4	F	9B/9HB(C-),9D/9HC(C-) 104A(C-), ‖115A	Thermo. and Stat. Mech.
	112L	1	F	40(C-), 45/ECS 36B/32C(C-) ‖112/‖CHE 110C	Comp. Lab in Stat. Mech.
	115A	4	F	9C/9HD(C-),9D/9HC(C-), 104A(C-), 105A(C-)	Quantum Mechanics I
	115B	4	W	104A(C-), 115A(C-)	Quantum Mechanics II
	115L	1	W	40(C-), 45/ECS 36B/32C(C-), 115A/CHE 110A, ‖115B*	Comp. Lab in Quantum Mech.
	115C*	4	S	40(C-), 115B, 45/ECS 36B/32C(C-)	Appl. of Quantum Mech.
				Either	
PHY	117	4	F	80(C-)	Phys. Instr. with A&D Elec.
	118	4	W	80(C-), 45/ECS 36B/32C(C-)	Phys. Instr. for Data Acq.
				or	
PHY	122A/B	4	WS	80(C-), 110B(C-) ‖112, 115A(C-)	Advanced Physics Lab

*: recommended, ‖: concurrently.

Electives: Additional electives to bring the total number of 3 or more unit upper division courses to 14, including at least three from capstone courses (Table 5), adding 4-5 courses, totaling 15-20 units with current offerings. Includes at most one from PHY 194H series, 195, 198, or 199 and excludes PHY 160.

Total Units (minimum): 110 (Table 3, Table 4, and electives $\implies 52 + 39 + 19$)

Table 5: Capstone Courses (BS Physics): Students choose two courses from one specialty and one course from a different specialty.
Units (minimum): 12

Course		Units	Offered	Prereqs	Name
				Nuclear and Particle:	
PHY	129A	4	S		Intro to Nuclear Physics
	130A	4	W		Elementary Particle Physics
	130B	4	S		Elementary Particle Physics
				Condensed Matter:	
PHY	140A	4	W		Intro to Solid State Physics
	140B	4	S		Intro to Solid State Physics
				Relativity and Astrophysics:	
PHY	154	4	S#	40,105B,110B,115A	Astro. Appl. of Phys.
	155	4	W	104A,105B,110A	General Relativity

Table 6: Math prerequisite for lower division courses related to this proposal. Where indicated, course grades are the minimum accepted to meet prerequisites. The prerequisites for the math courses are set by the math department and are only reported here. Where possible, math prerequisites for physics courses adopt the same choices made by the math department with respect to equivalent math coursework.

Course		Math Prereqs (Minimum Grade)	Name
MAT	17A		Calc (Bio&Med)
	17B	17A/19A/21A/21AH(C-)	Calc (Bio&Med)
	17C	17B(C-)	Calc (Bio&Med)
	21A		Calculus
	21B	21A/21AH(C-) or 17A/19A(B)	Calculus
	21C	21B/21BH/17C(C-) or 17B(B)	Calculus
	21D	21C/21CH(C-) or 17C(B)	Calculus
	22A	21C/21CH/17C(C-)	Linear Algebra
	22B	22A/27A/67(C-)	Diff. Eqs.
	27A	21C/21CH/17C(C-)	Linear Algebra
	27B	(22A+22AL)/27A/67(C-)	Diff. Eqs.
	67	21C/21CH(C-)	Mod. Lin. Alg.
PHY	9A	21B/21BH/21M/17C(C-) or 17B(B)	Class. Physics
	9B	21C/21CH(C-) or 17C(B)	Class. Physics
	9C	21D(C-)	Class. Physics
	9D	22A/27A/67(C-), ‖22B/‖27B(C-)	Class. Physics
PHY	9HA	‖21B/‖21BH/‖21M(C-)	Hon. Physics
	9HB	‖21C/‖21CH(C-)	Hon. Physics
	9HC	21C/21CH(C-)	Hon. Physics
	9HD	21D(C-)	Hon. Physics
PHY	45	‖22B/‖27B(C-)	Computational Physics

Table 7: An example schedule satisfying the *proposed requirements* for an undergraduate physics major that takes the 9H series and 122A. Physics course numbers are shown with the number of units in parenthesis. Math courses start with M. Courses in italics are electives or have at least two different offerings per year. Course CAP is a capstone elective, and course X is any advanced elective. Rate is 3-9 physics units per quarter in junior and senior year.

Year	Fall	Winter	Spring
Freshman	9HA(5) *M21B(4)*	9HB(5) *M21C(4)*	9HC(5) *M21D(4)*
Sophomore	9HD(5) *M22A(3)* 40(3)	105A(4) *M22B(3)* 45(4)	105B(4) *104A(4)* *80(4)*
Junior	115A(4) 112+L(5)	115B+L(5) 110A(4)	*115C(4)* 110B+L(5)
Senior	*X(3-4)*	*122A(4)* *CAP(4)*	*CAP(4)* *CAP(4)*

Table 8: An example schedule satisfying the *proposed requirements* for an undergraduate physics major that takes the 9 series and 122A. Physics course numbers are shown with the number of units in parenthesis. Math courses start with M. Courses in italics are electives or have at least two different offerings per year. Course CAP is a capstone elective, and course X is any advanced elective. Rate is 7-13 physics units per quarter in junior and senior year.

Year	Fall	Winter	Spring
Freshman	*M21A(4)*	*M21B(4)*	*M21C(4)* *9A(5)*
Sophomore	*M21D(4)* *9B(5)* 40(3)	*M22A(3)* *9C(5)*	*M22B(3)* *9D(4)* *80(4)*
Junior	*104A(4)* *X(3-4)*	105A(4) 110A(4) 45(4)	105B(4) 110B+L(5)
Senior	115A(4) 112+L(5)	115B+L(5) *122A(4)* *CAP(4)*	*115C(4)* *CAP(4)* *CAP(4)*

Table 9: An example schedule satisfying the *proposed requirements* for an undergraduate physics major that transfers to UC Davis in their junior year, without Physics 9D or 40 equivalents, and takes 122A. Physics course numbers are shown with the number of units in parenthesis. Courses in italics are electives or have at least two different offerings per year. Course CAP is a capstone elective, and course X is any advanced elective. Rate is 12-13 units per quarter. Note that only one upper division course is required in the first quarter of junior year, compared to three upper division courses in the current program.

Year	Fall	Winter	Spring
Junior	9D(4)	105A(4)	105B(4)
	40(3)	110A(4)	110B+L(5)
	104A(4)	45(4)	*80(4)*
Senior	115A(4)	115B+L(5)	*115C(4)*
	112+L(5)	*122A(4)*	*CAP(4)*
	X(3-4)	*CAP(4)*	*CAP(4)*

Several new courses have been added, some are no longer offered, and others require changes to their content:

- 9A-D and 9HA-D are unchanged, but their prerequisites have been adjusted for consistency with the math prerequisites (including minimum grades) used by the math department and to match existing policy (e.g. 9A is acceptable prerequisite for 9HB.)
- 9HE is no longer offered. This course is effectively an elective, with content that varies from instructor to instructor. By removing it, we allow the students in the honors sequence to start toward the core material sooner, leaving more time for advanced electives. With the more relaxed schedule in their senior year, it seems highly plausible that physics majors will take more advanced electives.
- 104A: This course will now be offered in both fall and spring, as discussed further in the discussion of prerequisites. Most students taking honors physics will take the course in the spring, while most transfer students will take the course in the fall. Two offerings will remove a major bottleneck, and allow for a smaller class sizes. Most four-year students from honors physics will take the spring offering, whereas most incoming junior year transfer students will take the fall offering. This will allow the course to be pitched slightly differently in each quarter, to better reflect student preparation.
- 105: The timing of the 105AB sequence is adjusted to start in winter. The content of 105AB should be at a level appropriate for a sophomore completing 9HD in the previous quarter. Mathematical physics (PHY 104A) is not a prerequisite, but typically students will take 104A either concurrently with 105B or before 105A.
- 110: The present curriculum devotes three quarters of required upper division coursework to Electricity and Magnetism (110ABC). This proposal eliminates 110C and increases the pace of 110AB. Students must reliably enter 110 having adequate preparation in vector calculus, curvilinear coordinates, Lorentz transformation, relativistic mechanics, and introductory electricity and magnetism. This material must be covered adequately in the 9 series and 104A.

- 112: The 115A prerequisite for 112 has been removed, and the treatment must rely on quantum from the PHY 9 series instead. Fermi and Bose statistics will be introduced independently in 112. PHY 9 and 9H must consistently cover discrete energy levels from the particle in a box and simple harmonic oscillator. The annual faculty curriculum discussion is the mechanism for ensuring this is so.

- The 115AB sequence is extended to include an elective third quarter: 115C. The prerequisites are 104A and 105A. The elective third quarter, 115C, adds 45 as a prerequisite, and the course includes extensive computational problems. The extra time should also allow coverage of new elective topics (for example Quantum Information Theory). As this is an elective course, there is no need to implement it immediately, and it can be developed first as a PHY 150 course.

- The computational courses are described in Section 5. PHY 40 is updated, PHY 102 and 104B are dropped, and new courses PHY 45, 110L, 112L and 115L are added. Computational Physics majors substitute ECS coursework for PHY 45, and the prerequisites of computational courses reflect this.

- The traditional lab courses are discussed in Section 6. PHY 80, 116A, 116B, and 116C are replaced with PHY 80, 117, and 118. PHY 122A/B is unchanged.

Example syllabi for all courses impacted by this proposal are presented in Section **??**.

This proposal is approximately neutral with respect to the cost of instruction if we assume that the cost of electives is held constant (we of course have no obligation to do so.) Instructors teaching a one-unit computation lab course will receive credit for one third of a standard podium course. The three new computational lab courses together add one new instructor. The second offering of 104A and the new required course 45 add two new instructors. But dropping courses 9HE, 102, 104B, 110C and replacing 116ABC with 117 and 118 frees 4.3 instructors, for a net decrease of 1.3 instructors. If the need for more sections of 80 requires an additional instructor (currently typically two, although three were initially planned for 2020-2021) there would be a decrease of 0.3 instructors. Additional fall offerings for 122A/B could be added as well, if needed to meet demand.

4 Theoretical Physics

At the core of any undergraduate physics degree are the following topics in theoretical physics:

- Classical Mechanics (105AB): the fundamental principles of physical laws (e.g. least action and symmetries) are taught in a familiar and intuitive context.

- Electromagnetism (110AB): a remarkable special case of classical phenomena that anticipate non-Newtonian physics (e.g. special relativity, gauge theories). No other force in nature can be understood so completely in such a straightforward fashion.

- Quantum Mechanics (115AB): the rules governing the microscopic world are different from those governing our familiar macroscopic world. The rules are not intuitive but they can be codified and used to make quantitative predictions which can be experimentally verified.

- Statistical Mechanics (112): the crucial statistical explanation for how microscopic laws ultimately produce the macroscopic world which we inhabit.

The physics department has recently adopted a procedure for maintaining example syllabi for core courses, through an annual faculty meeting discussion and vote. This is the mechanism for ensuring that prerequisite material is being appropriately and consistently covered in the proper courses. The example syllabi, as presented during the first annual faculty curriculum discussion are presented in Section **??**. This proposal need not resolve all of the issues brought up in these discussions, as these discussions will continue, with an update and faculty vote as needed.

The major topics from the first curriculum discussion were:

- It was agreed that 105A needs to cover Hamiltonian mechanics, by adding supplementary material to the preferred textbook (Morin).
- The treatment of radiation in 110AB is limited.
- There is support for covering waveguides in either 110AB, 80, or even 110L.
- PHY 104A plays a central role in providing students with the analytic techniques needed for upper division coursework. This course has been far too topical for the central role it plays in our program, and the example syllabi should be closely followed by future instructors.
- The 115AB content is presented as either the historical progression or spin first. The instructors will coordinate each year and agree on an approach. Sufficient review is included in both versions PHY 115B so that students who took 115A under a different progression do not miss any crucial material, such as particle-in-a-box, simple-harmonic oscillator, or angular momentum.

5 Computational Physics

One of the major objectives of this proposal is to better integrate computational physics throughout the curriculum. To do so, students must master a consistent set of tools that can be relied upon in later courses. The course descriptions in the course catalog will not reference specific tools, to allow this to evolve over time, but in a consistent manner. In this proposal, the supported computational tools are:

- Scientific computing tools for Python: NumPy, SciPy, MatPlotLib, and Jupyter Notebooks via Anaconda
- C/C++
- Computer Algebra: Mathematica or SymPy

The entire Scientific Python ecosystem is easily accessible to students for any major OS for free through Anaconda. No graphical features of C/C++ will be explored. If necessary, C++ programs will pass data by text file to SciPy for plotting. SymPy is not as mature as Mathematica, but comes with the significant advantages of being freely available within Anaconda and inter-operable with SciPy.

In 2018, we added a new required course, PHY 40, which introduces programming with both C++ and python. This format leaves no time for symbolic computation and little time for exploring additional features of Scientific Python beyond the basic Python language. In 2021, the course was taught exclusively with Scientific Python, which was found to be much more workable. However, even with this reduced scope, most students taking the course are not yet ready to do extensive

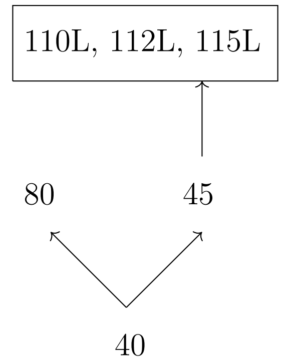

Figure 1: Prerequisite structure of computational courses.

unsupervised computational work, and only manage to make significant progress during lab sections with TA support. For this reason, recent instructors agree that a three unit course, with essentially no homework, is a more appropriate format for PHY 40.

In the current program, the additional computational physics requirement is either PHY 102 (1 unit) or 104B (4 units). This proposal replaces these options with a single new required four-unit computational physics course, PHY 45. In addition, BS physics majors are required to take three one-unit computational lab courses: 110L,112L, and 115L. PHY 80 is a traditional lab course, as described in Section 6, but it is included in this discussion as it also includes significant computational aspect.

The computational physics content of the required courses in this proposal include:

- 40: This three-unit course has no prerequisites and assumes no prior knowledge of programming. It provides an introduction to programming using examples from computational physics. It includes a short introduction to symbolic manipulations using a computer algebra system. The specific tools covered (which will not be included in the course description) are Scientific Python and either SymPy or Mathematica.

- 45: This new four-unit course, Computational Physics, will have the PHY 9 series through E&M as a prerequisite (9C/9HD) as well as PHY 40. The focus will be on solving physics problems at the conceptual level of the 9 series using computational physics. It will introduce the C++ programming language and continue using Scientific Python.

- 80: This traditional lab course includes extensive use of scientific python for data analysis and presentation, including curve fitting and plotting scientific data.

- 110L, 112L, and 115L: These three new one-unit computational lab courses are designed to be taken concurrently with 110B, 112, and 115B. They are computational problem solving labs related to E&M, statistical mechanics, and quantum mechanics. The emphasis will be on solving problems from upper division physics using computing at the level of PHY 45. One unit courses should involve three hours of academic work per week as described below. Each course is offered in a different quarter, so students get three units of computational problem-solving spread across at least one year. BS physics majors are required to take all three. A major emphasis of these lab courses is to give students practice at using programming to solve computational physics problems *independently*. They will be solving problems as homework, instead of in a lab section with TA support.

- 105B: this course now has PHY 40 as a prerequisite and can therefore include computational problems in mechanics using Scientific Python or computer algebra.
- 115C: This new elective course has PHY 45 as a prerequisite and is intended to include extensive computational problem solving as an integral part of the course.

Physics BS majors will be required to take 18 units (22 recommended) of coursework that involves extensive computing exercises. Furthermore, once student computing abilities are on more solid ground, capstone and advanced elective courses could further evolve to include additional computing exercises and add 45 (or at least 40) as a prerequisite. All majors now require PHY 40, so that could be added as a prerequisite to any upper division course. All applied majors include 45 or an equivalent, so, for example, 140A or 140B could include a computational component.

The content of PHY 40, 45, 105L, 110L. and 115L will likely evolve with our experience teaching these courses. Furthermore, the content of the one-unit computational labs will benefit from coordination with the lecturer for the corresponding lecture course (e.g. 115L and 115B). The annual faculty curriculum discussion will be one venue for these discussions to unfold.

One-unit workload: It is important that 110L, 112L, and 115L are taught as one-unit courses. A one-unit course should involve three hours of total academic work per week, which in this case includes the one hour of scheduled lecture time. For example, an appropriate workload would be five computing assignments due every two weeks during the ten week quarter. Each assignment would be introduced with a one-hour lecture, with a second one-hour lecture devoted to helping students complete the assignment. The assignments should take about five hours to complete, including one hour of help.

6 Experimental Physics

In traditional lab courses, students conduct scientific experiments, gain crucial hands on experience, and see theoretical concepts from a new perspective. The unit cap on required coursework places extreme time pressure on these essential lab experiences. In the current program, BS physics majors are only required to take four units of upper division traditional lab work, although many opt to take more.

Table 10: Content of New Lab Courses

Course	Replaces from	Content
80	116A	Passive Analog Electronics
	116C	Data Analysis
117	116A	Active Analog Electronics
	116B	Elementary Digital Electronics
118	116B	FPGAs
	116C	CPUs

PHY 80 was introduced in 2018, and is currently a prerequisite for 122A/B. It was anticipated that this would relieve some of the intense time pressure in these courses. The disruption from remote teaching during 2020 and 2021 has limited our ability to assess the effectiveness of this change, but preliminary indications were that incoming PHY 122A/B students were indeed better prepared.

In the current program, much of the content in PHY 80 is reproduced in the 116ABC sequence. PHY 116A covers passive analog electronics and basic lab equipment, and 116C covers data analysis

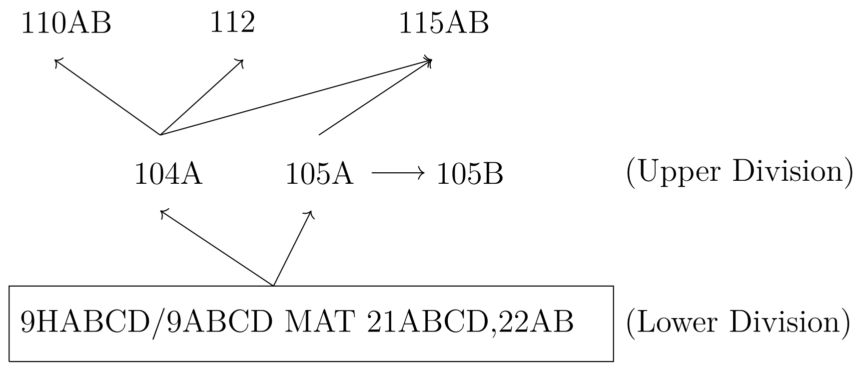

Figure 2: The prerequisite structure of required non-elective theoretical physics and math courses of the physics BS major. Internal prerequisite structure of lower division coursework is not shown. For clarity, only the most advanced prerequisites for each course are indicated.

including experimental uncertainties and curve fitting. Some students take PHY 80 before starting the PHY 116ABC sequence, while others do not. In this proposal, the three quarter 116ABC sequence is replaced with two new courses: 117, which covers analog and digital electronics, and 118, which covers data acquisition with microprocessors and FPGAs. Both courses have PHY 80 as a prerequisite. No content is lost in this remapping, which is summarized in Table 22

The 122A/B labs cycle students through experimental stations, and enrollment in each quarter is limited by the number of available stations. This proposal makes some adjustments to the prerequisites of 122A/B which would allow for fall offerings. This will help meet the expected increase in demand for 122A/B due to increased enrollment. As can be seen in Table 7, many seniors have room to take 122A/B in the fall of their senior year. We also plan to add additional stations to 122A/B, which is independent from this proposal.

PHY 80, 117, and 118 share the same dedicated lab space. The primary time for taking 80 will be in spring quarter, but it will also be offered in fall, and possibly winter, while 117 and 118 will be offered in fall and winter respectively. The department plans to expand the lab space available for 80, 117, and 118 to meet the needs of increased enrollment.

7 Prerequisites

An overview of the prerequisite structure of the core theoretical physics courses is shown in Fig. 2. PHY 104A is the major bottleneck in our program, as it requires the most advanced material from the first tier (MAT 22B) but is required for every course in the upper tier. The committee spent a great deal of time considering different ways to relieve or accommodate this bottleneck but in the end decided two offerings is the only effective way to achieve the goals of this proposal. This stems from the fact that incoming transfer students must start on 104A immediately upon arrival in fall to complete the remaining upper division courses in two years, but sophomores finishing 9HD in the fall are not generally ready to take 104A until the spring.

The most important changes to the prerequisite structure in the proposal are:

- The 9 and 9H prerequisites have been updated to better reflect the current policy actually enforced.
- PHY 112 does not require 115A anymore, it is taught at a level where 104A and PHY 9A-D

suffice.

- PHY 110A does not require 105A anymore.
- PHY 105B requires 40, so that computational examples can be used in this course if the instructor chooses.
- PHY 80 adds 40 as a prerequisite.
- PHY 122A/B adds a 115A prerequisite, which was implicit when 115A was a prerequisite for 112. Both 112 and 115A are allowed to be concurrent, although that will never happen unless we add a fall offering of 122A/B.
- The 116ABC sequence is replaced with independent courses 117 and 118 which each have 80 as a prerequisite.

The new prerequisite structure is considerably more flexible. In the junior year of Tables 8 and 9 missing or failing 104A or 105A jeopardizes a timely graduation, but instructor permission from one course (110A or 115A) will allow the student to proceed on schedule. The situation in Table 7 is even more forgiving.

8 Proposed BS with Specialization in Astrophysics

The astrophysics specialization requires updates to the prerequisites for PHY 151-158, mainly to accommodate the new schedule which offers PHY 105A in the winter and to add PHY 40 as a new required course. The astrophysics specialty courses now include 158, for a total of five, from which four are chosen. Limits on the number of units preclude including PHY 45, and the related lab courses, to the program. However, the PHY 151-158 courses already include extensive computational physics directly related to astrophysics, using the Python tool kit established in PHY 40.

The proposed required courses for the astrophysics specialization are presented in Table 3 (excluding PHY 45) and Table 11. To avoid version conflicts, some information in Table 4 is not duplicated here. PHY 108 has complicated prerequisites, as some non-majors from the PHY 7 series also take the course. The prerequisites are one year of introductory physics (PHY 9HA/9A plus 9B through 9D or 9HB through 9HD; or PHY 7ABC) and calculus (MAT 21A through MAT 21D). [**TODO: check if 9D really needed, and if MAT 17 is sufficient**] Example schedules are presented in Table 12.

Table 11: Depth Subject Matter: Astrophysics

Units (minimum): 60

Course		Units	Offered	Prereqs	Name
AST	25*	4			
PHY	104A	4			
	105A	4			
	108	3	S	(see text)	Optics
	108L	1	S	‖108	Optics Laboratory
	110A	4			
	110B	4			
	112	4			
	115A	4			
	115B	4			
PHY	157	4	S#	(same as 122A/B)	Astronomy Instrumentation and Data Analysis Lab
				or	
PHY	122A/B	4			Advanced Physics Laboratory
				Choose four of	
PHY	151	4	F#	‖40,‖104A	Stellar Structure and Evolution
	152	4	F#	‖40,‖104A	Galactic Structure and the Interstellar Medium
	153	4	W#	40,104A,‖105A	Extragalactic Astrophysics
	156	4	W#	40,104A,‖105A	Introduction to Cosmology
	158	4	S#	40,104A,105A	Galaxy Formation
				Choose two of	
PHY	105B	4			
	117	4			
	118	4			
	129A	4			
	130A	4			
	130B	4			
	150	4			Special Topics
	154	4			Astro. Appl. of Phys.
	155	4			General Relativity
GEL	163	4		AST 25, PHY 9ABC†	Planetary Geology and Geophysics
				at most one of	
PHY	194HAB	8			Special Study for Honors Students
	195	5			Senior Thesis
	198	3+			Directed Group Study
	199	3+			Special Study for Adv. Undergrads.

†: we have requested Geology update the prerequisites for GEL 163 to include 9H series.

*: recommended, #: not offered every year, ‖: concurrently.

PHY 108 prerequisites are complicated, and discussed in text. PHY 150 must be an astro topic and requires prior department approval. PHY 198 and 199 are assumed to be taken for 4 units in the unit tally.

Total Units (minimum): 108 (Table 3, no PHY 45, and Table 11 $\implies$ $52 - 4 + 60$)

Table 12: An example schedule for the junior and senior year of an astrophysics major. Physics course numbers are shown unless otherwise indicated. Courses in italics are electives. In most cases, the 9D requirement is satisfied before the junior year. The upper table starts with PHY 151 and the lower with 152.

Year	Fall	Winter	Spring
Junior	40 104A 151$^{\#}$ (9D)	105A 110A 153$^{\#}$	80 110B *105B/150*
Senior	115A 152$^{\#}$ 112	115B 156$^{\#}$ *130A/155/GEL 163*	108+L 157/122A/B *158$^{\#}$/129A/130B*

Year	Fall	Winter	Spring
Junior	40 104A 152$^{\#}$ (9D)	105A 110A 156$^{\#}$	80 110B *105B/158$^{\#}$*
Senior	115A 151$^{\#}$ 112	115B 153$^{\#}$ *130A/155/GEL 163*	108+L 157/122A/B *154$^{\#}$/129A/130B*

9 Proposed Applied Physics Majors

The applied physics majors overlap significantly with the physics major, and so most of the updates already proposed apply directly to the applied physics majors. In this proposal, all applied physics majors now require PHY 40, 45 (or an equivalent), 80, and at least one computing lab (from 110L,112L, and 115L). Most require at least two upper division lab courses. In some cases, the second lab course can be outside of physics.

Table 13: Depth subject matter for all applied physics majors excluding chemical physics.

Units: 24

Course		Units
PHY	104A	4
	105A	4
	110A	4
	110B	4
	112	4
	115A	4

The applied physics majors typically require significant upper division coursework outside of the physics department. This proposals makes several adjustments to the applied physics major requirements which remove required courses that are no longer offered regularly, or include a prohibitively large number of prerequisites. The proposed requirements take advantage of new relevant course offerings, by giving students more options to choose from, offering greater flexibility. In the tables, only the most relevant prerequisites are shown.

The applied physics majors require the preparatory courses in Table 3. However, the computational physics major does not require PHY 45, and instead requires significant computational coursework in computer science engineering (ECS).

For depth coursework, each applied physics major starts with a list of coursework common to all majors, in Table 13, and adds additional coursework specific to each specialization. The chemical physics majors are allowed to substitute quantum mechanics and thermodynamics coursework from the chemistry department, and so do not follow Table 13. This proposal includes modifications to all of the applied physics majors:

- The Computational Physics major requires either ECS 32 or ECS 36 sequence, reflecting student difficulty with registering for the latter sequence. The major does not require PHY 45 and ECS 36B or 32C will satisfy the prerequisite for the computing labs. The elective choices from the Computer Science and Engineering department have been extended. See Table 14.

- The Physical Electronics major excludes PHY 117 as part of the upper division lab requirement. The major instead includes a large amount of electronics from the Electrical and Computer Engineering department. Due to unit limits, only one computational lab is required. See Table 15.

- The Materials Science major is renamed Materials Physics, and the elective choices have been extended. Students may now use a Material Science and Engineering lab course as one of their required upper division lab courses. See Table 16.

- The Geological Physics major has been reworked to account for new prerequisites and infrequent offerings of courses like GEL 162. The major now features a choice between ATM 60 and GEO 60, both of which open up opportunities for upper division electives, and the elective choices have been expanded accordingly. The major features 6 units of coursework in Physics and Geology. See Table 17.

- In the past, the Atmospheric Physics major required the Physical and Chemical Oceanography course (GEL 150A), which now has extensive prerequisites that would exceed the unit limit. The Oceanography thrust of this course has been relocated to the electives, including the more easily obtained (GEL 116N). The list of Atmospheric Physics electives has been extended. See Table 18.

- The Physical and Chemical Oceanography course (GEL 150A) now has extensive prerequisites, which are now included explicitly as required courses. The elective choices have been updated. See Table 19. We presume (the more rigorous) CHE 4C is adequate for CHE 2C prerequisites despite not being listed in the course catalog.

- The Chemical Physics major requires the courses listed in Tables 3 and 20. Both chemistry and physics have significant lower division coursework as prerequisite for upper division coursework. The additional 15 units of lower division general chemistry course work required by this major limits the number of upper division courses that can be required. This major features two paths, one featuring PHY 112 and 115AB, and the other featuring CHE 110ABC. The latter choice relies on the treatment of quantum mechanics and thermodynamics provided by our colleagues in the chemistry department. This major requires 6 units of upper division lab work in chemistry or physics.

Examples schedules for Applied Physics majors are in Appendix A. Some of the majors might benefit from further revisions, which we plan to pursue once we have experience with the significant changes already proposed here. For example, it would be helpful if PHY 115AB were listed as an alternative prerequisite for CHE 110B. Or if PHY 40 could replace ECS 32A as prerequisite for ECS 32B.

Table 14: Computational Physics

Additional Preparatory Courses:
Units (minimum): 16 units.

Course		Units	Prereqs	Name
ECS	20	4	MAT 21A(C-)	Discrete Math for CS
			Either:	
ECS	32A†	4		Intro to Programming (Python)
	32B	4	ECS 32A/36A(C-)	Intro to Data Structures (Python)
	32C	4	ECS 32B(C-)	Impl of Data Structures (Unix/C)
	34*	4	ECS 34C(C-)	Software Dev. in Unix & C++
			or:	
ECS	36A	4		Programming and Problem Solving
	36B	4	ECS 36A(C-)	Software Dev. and OOP in C++
	36C	4	ECS 20(C-), 36B(C-)	Data Structs, Algs, and Programming

*: recommended, †: ECS 32AV can replace ECS 32A when the latter is unavailable. Students that don't place into ECS 36A may opt to start with ECS 32A.

Concentration Courses:
Units (minimum): 18 units. *: recommended, #: not offered every year, ‖: concurrently.

Course		Units	Prereqs	Name
			Choose two of:	
PHY	110L	1		
	112L	1		
	115L	1		
			Choose four of:	
			at least two of:	
ECS	117	4	ECS 20,32B/36C	Alg. for Data Science
	120	4	ECS 20,32B/36C	Theory of Computation
	122A	4	ECS 20,32B/36C	Algorithm Design & Analysis
	122B	4	ECS 122A, 36C	Algorithm Design & Analysis
	127	4	ECS 20,32A/32AV	Cryptography
	130	4	ECS 32A/32AV, MAT 22A	Scientific Computation
	132	4	ECS 34/36B,	Probability & Stat Modeling
	171	4	ECS 32B/36C, ECS 132, MAT 22A	Machine Learning
			and including:	
PHY	122A/B	4		
			or:	
PHY	117	4		
	118	4		

Additional Electives: (4 units) One course for at least 4 units of additional upper division coursework from ECS, MAT, or PHY.
Total Units (minimum): 110
(Tables 3, 13, and 14, no PHY 45, electives $\implies 52 + 24 + 16 + 18 - 4 + 4$)

Table 15: Physical Electronics

Additional Preparatory Courses:
Units (minimum): 4 units.

Course		Units	Prereqs	Name
ENG	17	4		Circuits I

Concentration Courses:
Units (minimum): 30 units.

Course		Units	Prereqs	Name
EEC	100	5	ENG 17	Circuits II
PHY	115B*	4		
	140A	4		
	140B*	4		
			Choose one of:	
PHY	110L	1		
	112L	1		
	115L	1		
			Choose one of:	
PHY	118	4		
	122A/B	4		
			Choose four of:	
EEC	110A	4	EEC 100, 140A‖	Electronic Circuits I
	110B	4	EEC 110A	Electronic Circuits II
	111	4	EEC 100	Digital Electronic Circuits
	140A	4	ENG 17	Principles of Device Physics I
	140B	4	EEC 140A	Principles of Device Physics II
	145	4	EEC 140A	Electronic Materials
	150	4	EEC 100, MAT 22AL	Intro. to Signals & Systems
	151	4	EEC 100	Digital Signals & Systems

*: recommended

Total Units (minimum): 110 (Tables 3, 13, and 15 $\Longrightarrow 52 + 24 + 4 + 30$)

Table 16: Materials Physics

Additional Preparatory Courses:
Units (minimum): 9 units.

Course		Units	Prereqs	Name
CHE	2A+2B	5+5		General Chemistry
	or			
CHE	4A	5		Gen. Chem. for Phys. Sci.
ENG	45	4	CHE 2B/4A(C-)	Properties of Materials

Concentration Courses:
Units (minimum): 25 units.

Course		Units	Prereqs	Name
PHY	115B	4		
	140A	4		
			Choose two of:	
	110L	1		
	112L	1		
	115L	1		
			Choose two† of:	
PHY	140B	4		
EMS	147	4	ENG 45	Princ. of Polymer Mat. Sci.
	160	4	ENG 45	Thermo of Materials.
	162	4	ENG 45	Struct. & Characterization
	170	4	ENG 45	Sustainable Energy
	172	4		Smart Materials
	174	4	ENG 45	Mech. Behavior of Materials
	180	4	ENG 45	Materials in Eng. Design
			Choose two of:	
PHY	122A/B	4		
	117	4		
	118	4		
			at most one of:	
EMS	162L	3	EMS 162‖	Structure & Characterization Lab
	170L	3	EMS 170‖	Sustainable Energy Lab
	172L	3	EMS 172‖	Smart Materials Lab
	174L	3	EMS 174‖	Mech. Behavior of Materials Lab

†: At least three, including PHY 140B, recommended.
Total Units (minimum): 110 (Tables 3, 13, and 16 $\implies 52 + 24 + 9 + 25$)

Table 17: Geological Physics

Additional Preparatory Courses:
Units (minimum): 9 units.

Course		Units	Prereqs	Name
G→EPS	50	3		Physical Geology
	50L	2		Physical Geology Laboratory
			Either:	
ATM	60	4		Intro. to Atmospheric Sci.
			or:	
CHE	4A†	5		Gen. Chem. for Phys. Sci.
	4B†	5	CHE 4A(C-)	Gen. Chem. for Phys. Sci.
G→EPS	60→101	4→3	CHE 4B, G→EPS 50+L	Earth Materials

†: CHE 2ABC can be substituted with consent of the department.

Concentration Courses:
Units (minimum): 11

Course		Units	Prereqs	Name
			Choose one† of:	
G→EPS	101→102#	3	GEL/EPS 50L	Structural Geology
	162#	3→4	GEL/EPS 50	Geophysics of the Solid Earth
	163#	3	GEL/EPS 50	Planetary Geology & Geophysics
EPS	164#	3		Seismology
			Choose two of:	
PHY	110L	1		
	112L	1		
	115L	1		
			At least 6 units:	
PHY	122A/B	4		
	117	4		
	118	4		
EPS	101L	2	EPS 101‖	Earth Materials Lab
G→EPS	101L→102L	2	G→EPS 50L,101→102	Structural Geology Lab

#: not offered every year, ‖: concurrently, † alternative upper division coursework from PHY, GEL, EPS, or ATM can be substituted, with consent of department, in case none of these courses are available.

Electives: Four additional upper division course offerings in ATM, GEL, or PHY of four or more units, or three units with consent of the department, or from the following list of pre-approved three-unit courses: EPS 101,102,109,116,132,137,

150B,161,162,163. Theoretical minimum is 12 units, but due to course scheduling, this is generally not possible to achieve, so the minimum units for this major will be capped at 108.

Total Units (minimum): 108 (Tables 3, 13, and 17 $\implies 52 + 24 + 9 + 11 + 12$)

Table 18: Atmospheric Physics

Additional Preparatory Courses:
Units (minimum): 7 units.

Course		Units	Prereqs	Name
G→EPS	50	3		Physical Geology
ATM	60	4		Intro. to Atmospheric Sci.

Concentration Courses:
Units (minimum): 24 units.

Course		Units	Prereqs	Name
ATM	120	4	ATM 60‖	Atmos. Thermo. and Cloud Physics
	121A	4	ATM 120	Atmospheric Dynamics
	121B	4	ATM 121A	Atmospheric Dynamics
			Choose two of:	
PHY	110L	1		
	112L	1		
	115L	1		
			Either:	
PHY	122A/B	4		
			or:	
PHY	117	4		
	118	4		
			Choose two of:	
PHY	105B	4		
	105C#	4		Continuum Mechanics
G→EPS	116N→116	3	GEL 50	Oceanography
	150A#	4	EPS 116,	Physical and Chemical Oceanography
	150B	3	GEL 50	Geological Oceanography
ATM	124	3	ATM 60	Meteorological Instruments & Observations
	128	4	ATM 60	Radiation and Satellite Meteorology
	158	4	ATM 121A	Boundary-Layer Meteorology

#: not offered every year, ‖: concurrently.

Elective: If PHY 122A/B lab option is chosen, one additional upper division course of 3 or more units, from PHY or ATM. **Total Units (minimum):** 110
(Tables 3, 13, 18, and elective $\implies 52 + 24 + 7 + 24 + 3$)

Table 19: Physical Oceanography

Additional Preparatory Courses:
Units (minimum): 18 units.

Course		Units	Prereqs	Name
CHE	4A†	5		Gen. Chem. for Phys. Sci.
	4B†	5	4A(C-)	Gen. Chem. for Phys. Sci.
	4C†	5	4B(C-)	Gen. Chem. for Phys. Sci.
G→EPS	50	3		Physical Geology
ATM	60*	4		Intro. to Atmospheric Sci.

*: recommended, †: CHE 2ABC can be substituted with consent of the department.

Concentration Courses:
Units (minimum): 16 units.

Course		Units	Prereqs	Name
G→EPS	116N→116	3	G→EPS 50	Oceanography
	150A†	4	GEL 116N, CHE 2C/4C	Phys. & Chem. Oceanography
			Choose two of	
PHY	110L	1		
	112L	1		
	115L	1		
			Choose one of	
PHY	122A/B	4		
	117	4		
	118	4		
			Choose one‡ of:	
PHY	105B	4		
	105C#	4		Continuum Mechanics
ATM	115	3	ATM 60	Hydroclimatology
	120	4	ATM 60‖	Atmos. Thermo. and Cloud Physics
G→EPS	108	3	GEL 50 CHE 2A/4A	Earth History: Paleoclimates
	109	3	GEL 50	Earth History: Sediments & Strata
	150B	3	GEL 50	Geological Oceanography

‖: concurrently, †: students should consult the EPS department course schedule to determine when EPS 150A will be offered, and plan accordingly, including completing EPS 116 beforehand. ‡: more than one is recommended if possible.

Total Units (minimum): 110 (Tables 3, 13, and 18 $\implies 52 + 24 + 18 + 16$)

Table 20: Chemical Physics

Additional Preparatory Courses:

Units (minimum): 15 units.

Course		Units	Prereqs	Name
CHE	4A†	5		Gen. Chem. for Phys. Sci.
CHE	4B†	5	4A(C-)	Gen. Chem. for Phys. Sci.
CHE	4C†	5	4B(C-)	Gen. Chem. for Phys. Sci.

*: recommended, †: CHE 2ABC can be substituted with consent of the department.

Concentration Courses:

Units 43 (minimum): units.

Course		Units	Prereqs	Name
PHY	104A	4		
	105A	4		
	110A	4		
	110B	4		
	110L*	4		
	112L*	4		
	115L*	4		
CHE	124A	3	CHE 4C	Inorganic Chemistry
	128A	3	CHE 4C	Organic Chemistry
			Either:	
PHY	115A	4		
	115B	4		
	112	4		
			or:	
CHE	110A	4	CHE 4C	Physical Chemistry: Intro to QM
	110B	4	CHE 110A	Physical Chemistry: Atoms and Molecules
	110C	4	CHE 110B	Physical Chemistry: Thermodynamics
			Choose one of:	
PHY	140A	4		
	140B	4		
CHE	124B	3	CHE 124A	Inorganic Chem
	124C	3	CHE 124A	Inorganic Chem
	128B	3	CHE 128A	Organic Chemistry
			Choose 6 or more units:	
PHY	122A/B	4		
	117	4		
	118	4		
CHE	105	4	CHE 110A	Analytical & Physical Chemical Methods
	115	4	CHE 105, 110B‖	Instrumental Analysis
	124L	2	CHE 124B/C	Laboratory Methods in Inorganic Chemistry
	129A	2	CHE 128A‖	Organic Chemistry Laboratory
	129B	2	CHE 129A,128B‖	Organic Chemistry Laboratory

*: recommended, #: not offered every year, ‖: concurrently.

Total Units (minimum): 110 (Tables 3 and 20 $\implies$ 52 + 15 + 43)

10 Proposed AB Physics Major

The proposed requirements for the AB Physics major are listed in Tables 3 (excluding PHY 45) and 21.

The college requirements for an AB degree include a cap of 80 units on major requirements. The current AB Physics major requires 82 units, which was approved by the Education Policy Committee in the past despite exceeding the cap. This proposal requires only 80 units for the AB, or which 32 units are in upper division coursework. Note that the university and college impose additional requirements beyond those discussed here.

Table 21: Depth Subject Matter (AB Physics)

Units (minimum): 32. *: recommended, ‖: concurrently.

Course		Units
PHY	104A	4
	105A	4
	110A	4
	110B	4
	112	4
	115A	4
PHY	122A/B	4
	At least one of	
PHY	129A	4
	130A	4
	140A	4
	151	4
	152	4
	153	4
	156	4
	158	4

Total Units (minimum): 80 (Tables 3, excluding PHY 45, and 21 $\implies$ $52 - 4 + 32$)

11 Implementation

Many aspects of this proposal involve changing the scheduling and number of offerings of certain courses (e.g. 104A, 115A), which does not require campus approval, and the department began implementing those changes in AY 2021-2022. We also proposed several new required classes, and all of these have been approved, added to the course catalog, and have been taught for several years now. These courses are PHY 45, 110L, 112L, 115L, 117, and 118. We have been encouraging majors to take advantage of the new course offerings, even though they are not yet required.

Changes to the curriculum are tracked in the integrated curriculum management system (ICMS). Each proposed change is approved at the level of UGCC/Vice-Chair, College, and then the Senate Committee on Courses of Instruction (COCI). Some of the changes proposed here, such as adjusting prerequisite requirements, do not directly depend on approval of the degree requirements, so we have pursued those as well.

Under these circumstances, we believe we can aggressively roll out the new requirements.

Table 22: Content of New Lab Courses

New Requirement	Replaces Requirement	Content
45(4)	102(1),104C(4)	Computational Physics
80(4), 117(4), 118(4)	116A(4),116B(4),116C(4)	Experimental Techniques
110A(4), 110B(4)	110A(4), 110B(4), 110C(4)	Electricity and Magnetism

There is a straightforward mapping of new course offerings to time slots in the current program. This should avoid most schedule conflicts. So for instance, the new winter offering of 105A will occupy the time slot for 105B in the current program. The mapping of new course offerings to time slots (by course) in the current program are worked out in Table 23. The remaining tables in this section are useful guides for students during the transition, see Tables **??**-**??**.

Table 23: Mapping of Course Schedule

Course	Current Course Time Slot
40 (fall)	102 (labs)
	105A (lecture)
45	104B
105A	105B
110AB	Unchanged
104A (spring)	110B
105B	115A
115A	105A
115B	105B
117	116A
118	116B

A Example Schedules for Applied Physics Majors

Examples schedules for each Applied Physics Major are provided in Tables 24-30. Only the junior and senior year are shown, as the previous two years are similar across all majors. Both community college (CC) and four-year students have the opportunity to complete most of their lower division coursework before their junior year, however, these examples only assume students have completed math and chemistry requirements and PHY 9A,9B, and 9C. Equivalents for those courses are readily available at CCs. Courses in the example schedules which have equivalents available at CCs (in assist.org) are indicated by parenthesis, indicating these can in principle be taken earlier than shown for many students. Courses in italics are electives. All of these example schedules were verified using recently available offerings, as posted on department websites or the course catalog.

As shown in the examples schedules, even with limited preparation at the CC, all of the majors can be completed in two years with at most 16 units per quarter. With more preparation at the CC, all of the majors can be completed in two years with at most 13 units per quarter.

Table 24: Example two-year schedule for a Computational Physics major.

Year	Fall	Winter	Spring
Junior	40 104A (ECS 32A)	105A 110A (ECS 20) (ECS 32B)	80 110B (9D) (ECS 32C)
Senior	*ECS 122A* 112+L 115A	*ECS 117* *ECS 120* 115L	*122A*

Table 25: Example two-year schedule for a Physical Electronics major. A particular feature of this major is the ability to take graduate courses in the senior year.

Year	Fall	Winter	Spring
Junior	40 104A (ENG 17)	45 105A 110A	80 EEC 100 110B (PHY 9)
Senior	112+L 115A *EEC 151*	140A *EEC 110A* *EEC 140A*	122A/B *EEC 110B* *EEC 245*

Table 26: Example two-year schedule for a Materials Physics major.

Year	Fall	Winter	Spring
Junior	40 104A (ENG 45)	45 110A 105A	80 110B+L (9D)
Senior	112+L 115A *EMS 180*	115B+L 122A 140A	*EMS 174* *EMS 174L*

Table 27: Example two-year schedule for a Geological Physics major.

Year	Fall	Winter	Spring
Junior	40 104A ATM 60 (GEL 50+L)	45 105A 110A	80 110B *GEL 161* (9D)
Senior	112+L 115A *ATM 120*	*118* *GEL 116N* *115B+L*	*GEL 109* *GEL 109L*

Table 28: Example two-year schedule for a Atmospheric Physics major.

Year	Fall	Winter	Spring
Junior	40 104A ATM 60	45 105A 110A (GEL 50)	80 *105B* 110B (9D)
Senior	ATM 120 112+L 115A	ATM 121A 115B+L	ATM 121B 122A *ATM 128*

Table 29: Example two-year schedule for a Physical Oceanography major.

Year	Fall	Winter	Spring
Junior	40 104A (GEL 50)	45 105A 110A	80 110B+L (9D)
Senior	112+L 115A	GEL 116N 118	GEL 150A *GEL 150B*

Table 30: Example two-year schedule for a Chemical Physics major.

Year	Fall	Winter	Spring
Junior	40 104A CHE 128A	45 110A 105A	80 110B (9D)
Senior	112 115A CHE 124A	115B *140A* *CHE 124B*	122A *124L*